CS24307 — THEORY (3.0 CR) | MODULE IV NOTES | BIT MESRA | NEP SCHEME

Module 4: Planning & Probabilistic Reasoning

STRIPS, PDDL, Graphplan Mutex, Bayes' Rule, Bayesian Belief Networks & d-separation

Module IV: Planning & Probabilistic Reasoning — Complete 7-Topic Syllabus Tracker

Topic 24: Planning in AI (Goal-Directed Action Sequences)

Topic 25: Components of Planning Problems (STRIPS Operators: PRE, ADD, DEL)

Topic 26: Types of Planning (State-Space, Partial-Order & Hierarchical HTN)

Topic 27: Goal Stack Planning (Subgoal Interactions & Sussman Anomaly)

Topic 28: Reasoning Under Uncertainty (Axioms of Probability)

Topic 29: Bayesian Inference (Prior, Likelihood, Posterior & Evidence)

Topic 30: Probabilistic Reasoning & Bayesian Belief Networks (DAGs)

Topic 24 & 25: Planning Foundations & STRIPS Representation

STRIPS Action Operator Representation (Fikes & Nilsson, 1971)

An action A is represented as a triple:

- **Action Name:** e.g., $\text{Move}(b, x, y)$ (Move block b from surface x to surface y).
- **Preconditions (PRE):** Set of positive logical literals that must be true for action to execute ($\text{On}(b, x) \wedge \text{Clear}(b) \wedge \text{Clear}(y)$).
- **Add-List (ADD):** Set of literals made true by executing the action ($\text{On}(b, y) \wedge \text{Clear}(x)$).
- **Delete-List (DEL):** Set of literals made false by executing the action ($\text{On}(b, x) \wedge \text{Clear}(y)$).

Topic 26 & 27: Goal Stack Planning & Types of Planning

Planning Paradigm	Operating Mechanism	Key Limitation / Advantage
1. Goal Stack Planning	Maintains a LIFO stack of subgoals and STRIPS operators. Solves subgoals sequentially one by one.	Vulnerable to **Subgoal Interleaving (Sussman Anomaly)**: achieving one subgoal may inadvertently undo a previously satisfied subgoal.
2. Partial-Order Planning (POP)	Works in plan space; adds actions and orders them only when strictly necessary (Least-Commitment principle).	Eliminates unnecessary search commitments; handles interleaved subgoals gracefully.
3. Hierarchical Task Network (HTN)	Recursively decomposes high-level abstract tasks into executable primitive action sequences.	Scales to industrial-scale engineering and manufacturing planning problems.

Topic 28 & 29: Reasoning Under Uncertainty & Bayesian Inference

1. Fundamental Axioms of Probability: - Range: $0 \leq P(A) \leq 1, \quad P(\text{True}) = 1, \quad P(\text{False}) = 0$ - Disjunction: $P(A \vee B) = P(A) + P(B) - P(A \wedge B)$ - Conditional Probability: $P(A \mid B) = \frac{P(A \wedge B)}{P(B)} \implies P(A \wedge B) = P(A \mid B)P(B)$

2. Bayes' Rule Formulation (Bayes, 1763):

$$P(\text{Hypothesis} \mid \text{Evidence}) = \frac{P(\text{Evidence} \mid \text{Hypothesis}) \cdot P(\text{Hypothesis})}{P(\text{Evidence})}$$

$$\text{Posterior} = \frac{\text{Likelihood} \times \text{Prior}}{\text{Evidence}}$$

🏛️ Step-by-Step Bayesian Medical Diagnosis Problem

A disease affects 1% of the population ($P(D) = 0.01$). A test has 95% true positive rate ($P(T \mid D) = 0.95$) and 5% false positive rate ($P(T \mid \neg D) = 0.05$). Find the probability a patient who tests positive actually has the disease $P(D \mid T)$.

1. Total Evidence $P(T) = P(T \mid D)P(D) + P(T \mid \neg D)P(\neg D) = (0.95)(0.01) + (0.05)(0.99) = 0.0095 + 0.0495 = \mathbf{0.0590}$.
2. $P(D \mid T) = \frac{P(T|D)P(D)}{P(T)} = \frac{0.0095}{0.0590} \approx \mathbf{0.161\ (16.1\%)}$.

Topic 30: Probabilistic Reasoning & Bayesian Belief Networks

Full Joint Distribution Chain Rule for Bayesian Networks: A **Bayesian Network** is a Directed Acyclic Graph (DAG) where nodes represent random variables and directed edges represent direct conditional dependencies:

$$P(X_1, X_2, \ldots, X_n) = \prod_{i=1}^{n} P(X_i \mid \text{Parents}(X_i))$$

Reduces storage from $O(2^n)$ in full joint table to $O(n \cdot 2^k)$ where k is maximum parent in-degree!

🧠 M4 Active Recall & Exam Questions

Q1. Define STRIPS. Explain the components of a planning action with a clear example. (8 Marks)

STRIPS (Stanford Research Institute Problem Solver) is a formal action representation language where world states are conjunctions of function-free positive literals.
An action is defined by: (1) *Name/Parameters*, (2) *Preconditions* (what must hold before execution), (3) *Add-List* (positive effects added to state), and (4) *Delete-List* (literals removed from state).
Example: $\text{Stack}(A, B)$
• $\text{PRE: Clear}(A), \text{Clear}(B), \text{On}(A, \text{Table})$
• $\text{ADD: On}(A, B)$
• $\text{DEL: Clear}(B), \text{On}(A, \text{Table})$